

Supporter & Patient Android Application

Comprehensive Codebase Dependencies Report

Target SDK Version: 33
Compile SDK Version: 34
Gradle Modules Analyzed: :app (Android Application), :data (Android Library)

This document lists, analyzes, and explains all external dependencies, compiler plugins, classpaths, and frameworks declared within the Support & Patient Android App workspace. The codebase is structured as a modularized project consisting of an main application module (:app) and a data persistence/networking module (:data). The following sections group dependencies logically to describe their unique engineering purposes.

1. Build Plugins & Classpaths (Build System)

These definitions govern the gradle execution ecosystem, compiler plugins, code generation pipelines, and external resource resolution classpaths.

Plugin ID / Classpath	Version	Modules Applied	Engineering Role / Description
com.android.application	8.3.2	:app	Configures the module as a buildable, runnable Android Application package (.apk/.aab).
com.android.library	8.1.0	:data	Configures the module as a reusable, modular Android Library (.aar).
org.jetbrains.kotlin.android	2.0.20	:app, :data	Integrates standard Kotlin language and compiler toolchain into Gradle build tasks.
kotlin-parcelize	Implicit	:app	Automates compiler-level Parcelable serialization code generation for Kotlin classes.
com.google.gms.google-services	4.4.0 (classpath) 4.3.15 (plugin)	:app	Extracts configuration values from google-services.json to bootstrap Firebase clients.

Plugin ID / Classpath	Version	Modules Applied	Engineering Role / Description
androidx.navigation.safeargs.kotlin	2.7.3 (classpath) Implicit (plugin)	:app	Generates safe, type-guaranteed direction classes to pass arguments across Navigation Graph views.
com.google.devtools.ksp	2.0.21-1.0.27	:app	Kotlin Symbol Processing framework. A lightweight compiler plug-in used for rapid annotation analysis and code generation (Room/Hilt).
com.google.dagger.hilt.android	2.51.1	:app	Configures compiler environments for Hilt Dependency Injection generation.
com.google.firebase.crashlytics	3.0.5	:app	Gradle hook to upload native symbols and obfuscation mapping files to the Firebase console.

2. Jetpack Core & AndroidX Libraries

Standard Jetpack libraries provided by Google to ensure predictable architecture, lifecycle awareness, backwards compatibility, and smooth visual UI component scaling.

Dependency Notation	Version	Module	Engineering Role / Description
androidx.core:core-ktx	1.9.0 (app) 1.17.0 (data)	:app, :data	Provides concise, idiomatic Kotlin extensions (KTX) for core platform API wrappers.
androidx.appcompat:appcompat	1.6.1 (app) 1.7.1 (data)	:app, :data	Delivers modern material styles, native ActionBars, and theme managers on legacy Android releases.
com.google.android.material:material	1.9.0 (app) 1.13.0 (data)	:app, :data	Renders highly visual components obeying Google's Material Design guidelines (widgets, cards, tabs).
androidx.constraintlayout:constraintlayout	2.1.4	:app	A powerful, flexible layout manager allowing flat, performant view arrangements without nested parent layouts.

Dependency Notation	Version	Module	Engineering Role / Description
androidx.activity:activity	1.9.0	:app	Exposes modern Android component hooks to manage layouts, UI overlays, and system back presses safely.
androidx.recyclerview:recyclerview	1.3.2	:app	Highly optimized scrollable container that recycles active Views for fast scroll performance with large data sets.
androidx.lifecycle:lifecycle-livedata-ktx	2.6.1	:app	Provides active lifecycle-aware wrappers around observable models, eliminating memory leaks and crashes.
androidx.lifecycle:lifecycle-viewmodel-ktx	2.6.1	:app	A state preservation engine protecting memory states across active configuration changes (e.g., orientation rotates).
androidx.lifecycle:lifecycle-viewmodel-android	2.8.3	:app	Core Android runtime integration layers matching lifecycle ViewModels.
androidx.navigation:navigation-fragment-ktx	2.7.1	:app	Decouples fragment transitions into declarative, graph-based configurations.
androidx.navigation:navigation-ui-ktx	2.7.1	:app	Integrates bottom menus, drawers, and actionbars with configured Navigation destinations.
androidx.work:work-runtime-ktx	2.9.0	:app	Manages guaranteed background execution tasks obeying power, internet, and battery constraints.
androidx.annotation:annotation	1.6.0	:app	Exposes structural flags checking thread constraints, asset resource types, and value limits.
androidx.legacy:legacy-support-v4	1.0.0	:app	Backwards compatibility hooks supporting legacy layout components and UI fragments.

3. Firebase Suite & Identity Providers

Handles the cloud-based serverless ecosystem including user accounts, firestore document models, realtime messaging synchronizations, file buckets, and notifications.

Dependency Notation	Version	Module	Engineering Role / Description
com.google.firebase:firebase-bom	33.11.0	:app	Firebase Bill of Materials (BoM). Ensures all declared Firebase libraries use aligned, fully-tested version combinations.
com.google.firebase:firebase-analytics	Managed	:app	Collects detailed telemetry and operational statistics representing user interactions.
com.google.firebase:firebase-auth-ktx	23.2.0	:app	Handles registration, sign-in, session validation, and JWT credentials securely in the cloud.
com.google.firebase:firebase-database-ktx	21.0.0	:app	A cloud-hosted, real-time NoSQL database keeping app and client values synced in milliseconds.
com.google.firebase:firebase-firestore-ktx	25.1.3	:app	Advanced scalable document database offering indexing, transactions, and robust offline queries.
com.google.firebase:firebase-storage-ktx	21.0.1	:app	Saves and retrieves heavy static files, user avatars, documents, and other multimedia formats.
com.google.firebase:firebase-messaging	24.1.1	:app	Manages downstream push notification signals and cloud messaging alerts.
com.google.firebase:firebase-inappmessaging-display	21.0.2	:app	Triggers high-impact dynamic slide-ins, updates, and custom modal prompts on client screens.
com.google.firebase:firebase-crashlytics-ndk	Managed	:app	Captures native C++ architecture crashes for deep stability investigations.
com.firebaseui:firebase-ui-auth	8.0.2	:app	Generates pre-optimized, ready-to-use login flows and visual user registration dialogs.
com.firebaseui:firebase-ui-database	8.0.2	:app	Binds live Firebase DB query nodes directly to recyclerview adapters for auto-refreshing feeds.

4. Local Cache & Persistence (Room)

Abstracts low-level SQLite configurations to provide type-safe, compile-time verified local queries and offline state persistence.

Dependency Notation	Version	Module	Engineering Role / Description
androidx.room:room-runtime	2.6.1	:app	Manages the local database schema lifecycle, connections, and statement execution managers.
androidx.room:room-ktx	2.6.1	:app	Provides coroutines and flow bindings to observe dynamic, live local DB queries.
androidx.room:room-compiler	2.6.1 (KSP)	:app	Annotation analyzer. Translates @Entity, @Dao, and @Query statements into optimized SQL compilation scripts.

5. Dependency Injection (Dagger Hilt)

Decouples class instantiations, enforces solid architecture, and simplifies code testability by automatically inserting classes at runtime.

Dependency Notation	Version	Module	Engineering Role / Description
com.google.dagger:hilt-android	2.51.1	:app	Preconfigured dependency injector engine specializing in modular, clean Android component lifetimes.
com.google.dagger:hilt-compiler	2.51.1 (KSP)	:app	Build-time generator building dependency factories, scoping layers, and component bindings.

6. Media, Utilities, Networking & Visual Extras

A suite of third-party libraries for media streaming, network operations, asynchronous tasks, image loading, and specific UI calendars/carousel indicators.

Dependency Notation	Version	Module	Engineering Role / Description
org.jetbrains.kotlinx:kotlinx-coroutines-core	1.7.3	:app	Core API supporting light threads (coroutines) for highly asynchronous execution structures.

Dependency Notation	Version	Module	Engineering Role / Description
org.jetbrains.kotlin:kotlinx-coroutines-android	1.7.3	:app	Exposes Android-specific UI thread dispatchers, keeping operations fluid and non-blocking.
com.google.android.exoplayer:exoplayer	2.16.1	:app	High-performance modular player for video/audio streams, supporting MP4, HLS, DASH, and offline media playback.
com.github.bumptech.glide:glide	4.16.0	:app	Intelligent media cache and downsampler. Automatically loads, formats, and displays remote images with visual optimizations.
com.google.code.gson:gson	2.8.6	:app	Parser library mapping Kotlin objects into standard JSON representation strings and vice-versa.
com.squareup.okhttp3:okhttp	4.12.0	:app	An optimized HTTP client wrapping connection pools, header managers, redirect caches, and socket engines.
com.prolificinteractive:material-calendarview	1.4.3	:app	A customizable calendar view layout showing schedule selectors, events, and monthly timelines.
com.tbuonomo:dotsindicator	4.3	:app	Small stylish layout displaying dot carousels matching sliding screens and dynamic page viewers.

7. Quality Assurance & Automated Testing

These dependencies are loaded only during testing configurations to facilitate local unit testing and on-device instrumented UI automated tests.

Dependency Notation	Version	Module	Test Scope	Engineering Role / Description
junit:junit	4.13.2	:app, :data	test (Unit)	Core offline JUnit assertion framework for testing classes, logic, and helpers in isolation.

Dependency Notation	Version	Module	Test Scope	Engineering Role / Description
com.google.truth:truth	1.4.1	:app	test (Unit)	Google's assertion library providing exceptionally clear, readable test error messages.
androidx.test.ext:junit	1.1.5 (app) 1.3.0 (data)	:app, :data	androidTest (UI)	Integrates standard JUnit test execution with Android instrumentation APIs.
androidx.test.espresso:espresso-core	3.5.1 (app) 3.7.0 (data)	:app, :data	androidTest (UI)	Automates real user gestures (clicks, sweeps, keystrokes) to test UI flow on virtual/real devices.
androidx.test:core-ktx	1.5.0	:app	androidTest (UI)	Provides clean, declarative Kotlin extension helper methods for instrumentation runners.

```

├── app/
|   ├── MyFirebaseMessagingService.kt
|   ├── data/
|   |   └── AppDatabase.kt

```

```
| |— di/
| |  └─ AppModule.kt
|  └─ presentation/
|    |— AppViewModel.kt
|    |— MyApplication.kt
|    └─ SplashActivity.kt
|— auth/
|  |— data/
|  |  |— entity/
|  |  |  |— Partner.kt
|  |  |  └─ UserProfile.kt
|  |  └─ repo/
|  |    └─ AuthRepo.kt
|  |— di/
|  |  └─ Di.kt
|  └─ presentation/
|    |— AgeFragment.kt
|    |— AuthActivity.kt
|    |— AuthViewModel.kt
|    |— GenderFragment.kt
|    |— LanguageFragment.kt
|    |— LoginFragment.kt
|    |— MainAuthFragment.kt
|    └─ SignUpFragment.kt
```


- | └─ VerificationEmailFragment.kt
- | └─ base/
 - | └─ BaseActivity.kt
 - | └─ BaseBottomSheetDialogFragment.kt
 - | └─ BaseDialogFragment.kt
 - | └─ BaseFragment.kt
 - | └─ BaseViewModel.kt
 - | └─ LisetnerViewModel.kt
- | └─ chatting/
 - | └─ data/
 - | | └─ entity/
 - | | | └─ Chatting.kt
 - | | | └─ Message.kt
 - | | | └─ MetaDataMessages.kt
 - | | └─ repo/
 - | | └─ ChattingRepo.kt
 - | └─ di/
 - | | └─ DI.kt
- | └─ presintation/
 - | └─ ChatActivity.kt
 - | └─ ChatFragment.kt
 - | └─ ChatViewModel.kt
 - | └─ ChooseUserToChatFragment.kt
 - | └─ MessagesAdapter.kt

- | |─ MessagesListAdapter.kt
- | |─ MessagesListFragment.kt
- | |─ SupportersChattingAdapter.kt
- | |─ getLibraryContent/
 - | |─ data/
 - | |─ DepressionMisconception.kt
 - | |─ DepressionMisconceptionRepo.kt
 - | |─ LibraryContent.kt
 - | |─ LibraryContentRepo.kt
 - | |─ LibraryDao.kt
 - | |─ di/
 - | |─ LibraryDi.kt
- | |─ presentaion/
 - | |─ ArticleFragment.kt
 - | |─ CommonContentFragment.kt
 - | |─ CommonContentLibraryAdapter.kt
 - | |─ CustomizedContentFragment.kt
 - | |─ CustomizedContentLibraryAdapter.kt
 - | |─ LibraryActivity.kt
 - | |─ LibraryContentFragment.kt
 - | |─ LibraryViewModel.kt
 - | |─ MisconceptionAdapter.kt
 - | |─ MisconceptionsAboutDepressionFragment.kt
 - | |─ SuggestedContentFragment.kt

|— interfaces/

| |— DataReLoader.kt

| |— ItemMessagesListClicked.kt

| |— ItemPostsClicked.kt

| |— ItemStepsClicked.kt

| |— ItemSupplicationClicked.kt

| |— OnConfirmButtonClicked.kt

| |— OnItemLibraryContentClicked.kt

|— model/

| |— CurrentTask2.kt

| |— DataPatient.kt

| |— DayTask.kt

| |— Duration.kt

| |— Partner.kt

| |— Permissions.kt

| |— StatusDailyProgram2.kt

| |— Step.kt

| |— Supplication.kt

| |— SupplicationsUser.kt

| |— Task.kt

| |— TestD.kt

|— notifications/

| |— DI.kt

| |— data/

```
| | └─ entities/
| | |   └─ Notification.kt
| |   └─ repos/
| |     └─ NotificationsRepo.kt
|   └─ presentation/
|     └─ NotificationsAdapter.kt
|     └─ NotificationsFragment.kt
|     └─ NotificationsViewModel.kt
└─ numbersHelping/
    └─ HelpingAdapterAdapter.kt
    └─ NumberHelpingFragment.kt
└─ posts/
    └─ OnSendButtonClicked.kt
    └─ data/
        └─ entity/
            └─ Post.kt
            └─ Posts.kt
            └─ repo/
                └─ PostsRepo.kt
            └─ di/
                └─ DI.kt
            └─ presentation/
                └─ ChooseSupporterFragment.kt
                └─ ChooseSupportersAdapter.kt
```

- | |─ DisplayArticleFragment.kt
- | |─ DisplayGratitudeFragment.kt
- | |─ DisplaySupplicationFragment.kt
- | |─ PostsActivity.kt
- | |─ PostsAdapter.kt
- | |─ PostsFragment.kt
- | |─ PostsViewModel.kt
- |─ users/
 - | |─ data/
 - | | |─ entity/
 - | | | |─ InvitationCode.kt
 - | | | |─ ReligionData.kt
 - | | | |─ repo/
 - | | | | |─ UsersRepo.kt
 - | | |─ DI.kt
 - | |─ presentation/
 - | | |─ EditProfileActivity.kt
 - | | |─ EditProfileViewModel.kt
 - | | |─ FragmentAddSupporter.kt
 - | | |─ ProfileFragment.kt
 - | | |─ SettingsFragment.kt
 - | | |─ SupporterDetailsFragment.kt
 - | | |─ SupporterHomeFragment.kt

| └─ SupportersFragment.kt

| └─ UserHomeFragment.kt

└─ util/

└─ ClickShrinkEffect.kt

└─ ConnectionReceiver.kt

└─ Constant.kt

└─ CustomToast.kt

└─ Extension.kt

└─ Resource.kt

└─ SharedPreferencesManager.kt

1.2 Database Schema

1.2.1 Firestore Database Schema

1.2.1.1 Daily Program Schema

- **Collection:** daily_program
 - **Document:** 1 (e.g., day-1)
 - **Field:** behaviorActivation (Array)
 - **Item (Map)**
 - ageGroup : Number
 - arDescription : String (Arabic)
 - arTitle : String (Arabic)
 - enDescription : String (English)
 - enTitle : String (English)
 - gender : Number
 - id : String
 - image : String
 - link : String
 - religion : Boolean
 - type : Number
 - **Field:** educational (Array)
 - **Item (Map)**
 - ageGroup : Number
 - arDescription : String (Arabic)
 - arTitle : String (Arabic)
 - enDescription : String (English)
 - enTitle : String (English)
 - gender : Number
 - id : Null
 - image : String
 - link : Null
 - religion : Boolean
 - type : Number
 - **Field:** spiritual (Array)
 - **Item (Map)**

- ageGroup : Number
- arDescription : String (Arabic)
- arTitle : String (Arabic)
- enDescription : String (English)
- enTitle : String (English)
- gender : Number
- id : Null
- image : String
- link : Null
- religion : Boolean
- type : Number

1.2.1.2 Supplications Schema

- **Collection:** supplications

User-created Supplications

- **Document:** user-id
 - Array Items (Map)
 - name : String (Supplication content)
 - number : Number (Number of repetitions)

Application Supplications

- **Document:** app
 - Array Items (Map)
 - name : String (Supplication content)
 - number : Number (Number of repetitions)

1.2.1.3 User Profile Schema

- **Collection:** profiles_users
 - **Document:** userId_1 (e.g., user12345)
 - email : String
 - id : String
 - name : String

- partnerEmail : String
- partnerId : String
- imageUser : String
- gender : Number
- ageGroup : Number
- religion : Boolean
- token : String
- invitationUsed : Boolean
- invitationBase : String
- invitationCode : String
- supportersNumber : String
- hasPartner : Boolean
- status : Number
- currentDay : String
- allowDailyProgramDetails : Boolean
- allowMoodTrackingDetails : Boolean
- allowPrivateMessages : Boolean

1.2.1.4 Posts Schema

- **Collection:** posts
 - **Document:** email_1 (e.g., ali@gmail.com)
 - **Field:** posts (Array)
 - **Item (Map)**
 - postIndex : Number
 - timeStamp : DateTime
 - type : String
 - supports (Map)
 - email : String
 - Multiple supporter emails can be stored.
 - seen_list (Map)
 - email : String
 - Multiple viewer emails can be stored.
 - gratitude (Map)
 - answer : String
 - index : String
 - timeStamp : DateTime

1.2.1.5 Images Schema

- **Collection:** images
 - **Document:** user_id
 - imageUrl : String

1.2.1.6 Chats Schema

- **Collection:** chats
 - **Document:** user_id

Messages

- messages (Array)
 - **Item (Map)**
 - seenByCaregiver : Boolean
 - seenByPatient : Boolean
 - senderId : String
 - timeStamp : String
 - text : String

Metadata

- meta (Object)
 - idCaregiver : String
 - idPatient : String
 - imageCaregiver : String
 - imagePatient : String
 - nameCaregiver : String
 - namePatient : String

1.2.2 Realtime Database Schema

1.2.2.1 Library Content Schema

- **Root Node:** library_contents
 - **Node:** id (e.g., 0)

- `id : Integer`
- `date : String`
- `category : String`
- `backgroundColor : String`
- `enText : String`
- `arDescription : String`
- `enDescription : String`
- `arTitle : String`
- `enTitle : String`
- `subCategory : String`
- `type : String`
- `duration : String`
- `imageUrl : String`
- `videoURL : String`
- `audioURL : String`
- `rate : Integer`
- `viewCount : Integer`
- `religion : Boolean`

1.2.3 Room Database Schema

1.2.3.1 Task Content Schema

Task Entity

Field	Type	Example
day	String	"1"
nameTask	String	"task1"
image	Integer	Image resource ID
isCompleted	Boolean	true
description	String	"desc task1"